MeshSmith

text or image → **print-ready + game-ready** 3D assets · fully local

AMD AI DevMaster Hackathon
Track 1 · Multimodal Content Creation
runs on Radeon · ROCm

```
prompt ─► SDXL-Turbo ─► u2net cutout ─► Hunyuan3D-2 DiT ─► repair (watertight) ─┬─► decimate ─► xatlas UV ─► PBR ─► GLB / OBJ   (game)
photo ──────────────────────────────►┘   all inference local, fp16, torch.cuda API   └─► scale-to-mm ─► 26-orientation support search ─► wall-check ─► STL + report  (print)
```

THE GAP IT CLOSES

Image→3D models make impressive meshes that **fail at the point of use**: not watertight, no UVs, no physical scale, floating debris, orientations that drown FDM prints in support material. MeshSmith is the missing last mile — from "look at this mesh" to **"here is the STL your slicer accepts."**

WHO IT'S FOR

- **3D-printing hobbyists** — prompt → knob, bracket, figurine. No CAD.
- **Indie game devs** — props with sane topology, UVs, PBR materials.
- **Makerspaces & schools** — no API fees, data stays on-device, and the report *teaches* printability.

0
cloud API calls in the core pipeline

3
open models chained on one GPU

2
finished artifacts per prompt

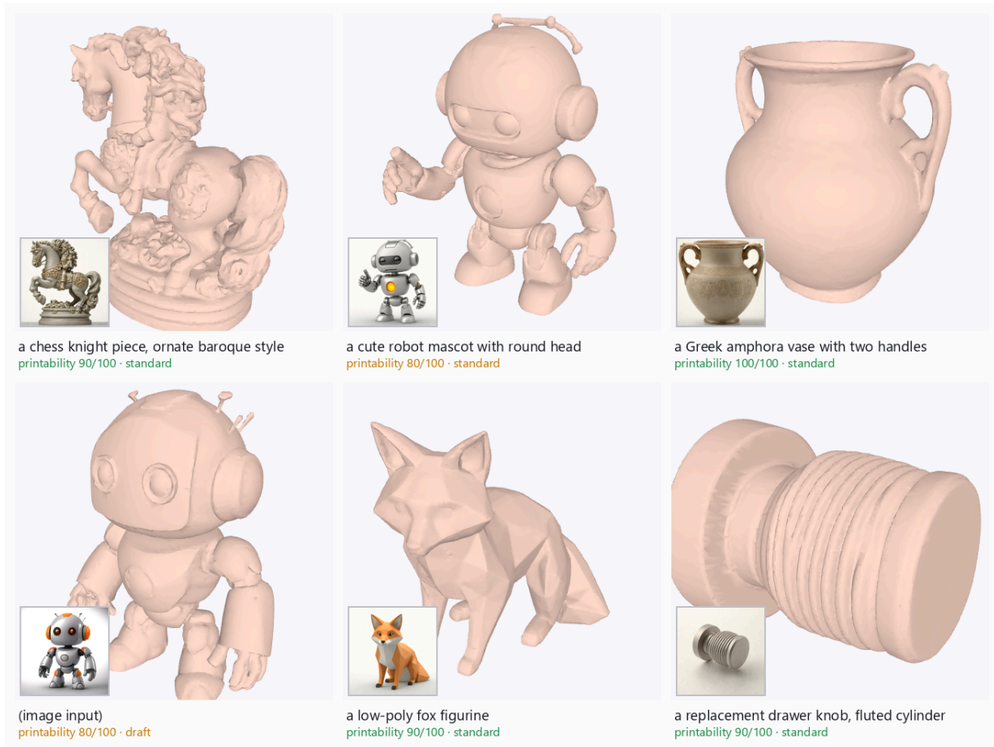
BUILT FOR ROCM, NOT PORTED TO IT

- **Zero vendor-specific code** — GPU stages use the portable `torch.cuda` API; ROCm maps it to HIP.
- **Shape-only Hunyuan3D by design** — skips the custom CUDA texture rasterizer, so install is pure `pip` in any `rocm/pytorch` container.
- **fp16 + lazy load/unload** — fits 16 GB cards.
- **ROCm Dockerfile** + one-command Radeon Cloud setup.
- **Observable** — per-stage wall time & peak VRAM in every report.

MEASURED — WARM, STANDARD

PROMPT	3D S	TOTAL S	SCORE
drawer knob, fluted	72.4	130.2	90
low-poly fox figurine	72.1	129.2	90
Greek amphora vase	76.7	181.3	100

`draft` preset (mini-turbo DiT): **~25 s/asset** warm. Across all five benchmark runs auto-orientation cut support volume by **0–82%** (0% = the amphora, already upright — the search correctly leaves good orientations alone).



Inset = the generated concept image, large = the finished mesh. Organic figures, hard-surface parts, vessels — and bottom-left, a photo taken straight to 3D. **Every asset ships with a printability report:** watertight ✓ · auto-orientation (support saved %) · wall thickness min/p05 · bed fit · score /100.

```
pip install -r requirements.txt
python app.py  # Gradio UI + 3D viewer
python -m meshsmith.cli "a chess knight" --quality high
```